Jinha Choi

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Research Interests

Physics-informed Dexterous Manipulation, Humanoid Learning, Reinforcement Learning

Education

Yonsei University, Seoul, South Korea *Mar. 2021 – Feb. 2027 (Expected)*
B.S. Candidate in Computer Science *(Military Service: Mar. 2023 – Sep. 2024)*
B.S. Candidate in Nano Science and Engineering GPA: 3.89 / 4.3

West Vancouver Secondary School, British Columbia, Canada *Sep. 2014 – Jun. 2020*
International Baccalaureate Diploma

Research

Yonsei RoPhiLab, Seoul, South Korea | *Prof. Yonghyeon Lee* *Sep. 2026 – Present*

Developing high-DoF dexterous manipulation methods with physics-informed priors to improve adaptability and generalization in unstructured environments.

KAIST HuGeLab (Summer Intern), Seoul, South Korea | *Prof. Beomjoon Kim* *Jun. 2026 – Aug. 2026*

Developed and deployed learning-based locomotion policies on the Unitree G1 and a custom 108 kg humanoid, with an emphasis on robustness across simulated and real-world conditions.

Yonsei DILLAB, Seoul, South Korea | *Prof. Jongmin Lee* *Jan. 2026 – Jun. 2026*

Published a paper [C1] on an offline-RL method that distills a value-aware multi-step flow policy into an efficient one-step policy using state-wise entropic optimal transport, improving multimodal policy learning across benchmark tasks.

Yonsei RLLAB, Seoul, South Korea | *Prof. Youngwoon Lee* *Jul. 2025 – Dec. 2025*

Built humanoid locomotion baselines using retargeted motion priors. Analyzed how retargeting choices affect motion quality and stability, and explored reusable motion components with less task-specific reward engineering.

Publications

[C1] J. Shon*, **J. Choi*** J. Jeon, and J. Lee, "Learning Multimodal One-step Flow Policy via Value-weighted Optimal Transport," *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.
*Equal contribution.

[P1] G. Son, J. Hong, H. Fan, H. Nam, H. Ko, S. Lim, J. Song, **J. Choi**, G. Paulo, Y. Yu, and S. Biderman, "When AI Co-Scientists Fail: SPOT—a Benchmark for Automated Verification of Scientific Research," (2025). arXiv preprint.

Experience & Projects

Grasping via Global Planning and Local Correction | *Prof. Yonghyeon Lee* *Oct. 2026 – Present*

- Developing a hierarchical grasping system coupling real-time global planning and control based on robot kinematics and dynamics with learning-based local correction.
- Integrating side-view and wrist-mounted visual feedback across both control levels, targeting robust grasping across diverse objects and positions.

Franka FR3 Control and Perception | *Prof. Yonghyeon Lee* *Sep. 2026 – Oct. 2026*

- Implemented torque-level controllers on a Franka FR3, including gravity compensation, collision-aware control, QP-based inverse kinematics, and joint/task-space impedance control.
- Integrated a calibrated two-camera setup with point-cloud perception and visual servoing for perception-guided grasping on the physical robot.

Humanoid Deployment – Locomotion | *Prof. Beomjoon Kim* *Jul. 2026*

- Trained, validated, and deployed locomotion policies on the Unitree G1 and a custom 108 kg humanoid via sim-to-sim and sim-to-real evaluation.

Franka FR3 Teleoperation Data Collection | *Prof. Beomjoon Kim* *Jul. 2026*

- Integrated a Franka Panda with a 3D mouse, Touch haptic device, and Meta Quest for teleoperated demonstration collection.

Humanoid Fixed-Base Manipulation | *Prof. Youngwoon Lee* *Mar. 2026 – Jun. 2026*

- Investigated PPO exploration limits for humanoid fixed-base manipulation in massively parallel simulation.
- Compared PPO with SAPG and EPO and analyzed methods for improving exploration.

A1-Challenge Autonomous Driving Competition | *Ministry of Trade & KIAPI* *Mar. 2026 – May 2026*

- Placed 4th; qualified in Nov. 2025.
- Built behavioral-cloning and DAgger racing policies in MORAI using targeted corner-case demonstrations.
- Explored reinforcement learning to improve robustness and driving performance.

Physical AI Robotics Hackathon – Arm Manipulation | *Kernel Academy & Doosan Robotics* *Mar. 2026*

- Built a speech-guided pick-and-place pipeline with a Doosan E0509, RGB-D sensing, Whisper, SAM2, GPT-4o, and ROS2.
- Implemented calibrated 3D localization, task verification, and error recovery for autonomous execution.

Image Inpainting with LLMs and Diffusion Models | *OUTTA AI Bootcamp* *Jul. 2024 – Aug. 2024*

- Developed a language-conditioned image editing pipeline combining LLM-based attribute extraction, semantic segmentation, and diffusion-based inpainting.

Korea Defense Intelligence Command – Translator | *Military Service* *Mar. 2023 – Sep. 2024*

- Translated classified documents and interpreted classified meetings for KDIC and partner agencies, including the National Intelligence Service and the U.S. National Geospatial-Intelligence Agency.

VEX Robotics | *Team Lead, 1010B, WVSS Robotics* *Sep. 2017 – Jun. 2020*

- Led a 12-member team across three seasons, overseeing strategy, mechanical and electrical design, programming, autonomous control, and competitive driving.
- Earned first-place finishes at regional and British Columbia provincial competitions and advanced to the VEX Robotics World Championship.

Selected Talks & Presentations

Learning Multimodal One-step Flow Policy via Value-weighted Optimal Transport *Jul. 2026*

Research Talk, YAI (Yonsei University AI Club)

Exploration Behavior of PPO in Massively Parallel Reinforcement Learning *Jun. 2026*

Poster Presentation, Yonsei UIC Science Fair

Awards & Honors

Hyundai NGV Robotics Track H-Mobility Class Certificate *Mar. 2026 – Jul. 2026*

Completed Hyundai's Robotics Track in robotics and robot learning and received the H-Mobility Class Certificate.

2nd Place, Physical AI Robotics Hackathon *Mar. 2026*

Presented a manipulation pipeline and demonstrated perception-guided grasping at the Kernel Academy hackathon.

Top Award, Korea Defense Intelligence Command AI Security Competition *2024*

Awarded by Major General Sangho Moon (OF-7), Commander, KDIC.
Proposed an on-device AI framework for military systems with partial internet exposure, designed to support operational convenience while strengthening protection against security vulnerabilities and malicious misuse.

Honor Roll, Yonsei University *Jul. 2022*

Recognized for academic performance in the top 3%.

1st Place, Yonsei–Nexon RC Creative Platform *Mar. 2021 – Dec. 2021*

Led a five-member team to develop an automated revolving-door assistance system for people with mobility impairments; ranked 1st of 90 teams and received a $10,000 award.

Division Finalist, VEX Robotics World Championship – Kentucky, USA *Nov. 2019*

Reached the Science Division final in a 100-team division within a field of 600 qualified teams.

1st Place, VEX Robotics Canada BC Provincial Championship *Jun. 2019*

Won the British Columbia Provincial Championship and qualified for the VEX Robotics World Championship.